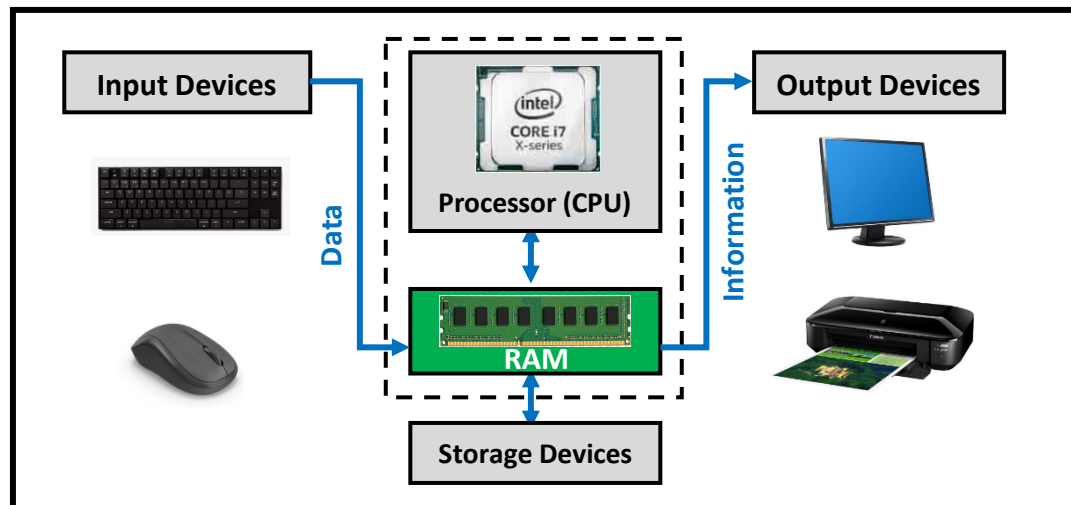


Lec 1: COMPUTER FUNDAMENTALS- Part1

1.1 BASICS OF A COMPUTER

❖ What is a computer?

A **computer** is an electronic machine that accepts data, stores and processes it according to instructions (programs) in order to produce the results (information). The word “computer” comes from the word ‘compute’, which means to calculate.



Data: is a raw material and unorganized facts that need to be processed.

Information: When data is processed, organized, structured or presented in a given context to make them useful, it is called **Information**.

1.2 CLASSIFICATIONS OF COMPUTERS

1.2.1 Personal Computer (or Microcomputer)

- **Desktop Computer:** a personal computer designed to fit on a desk. It is typically made up of a system unit (Case), a keyboard, a mouse, and a monitor.

- **Laptop Computer:** a portable computer with an integrated screen and keyboard. It is battery-powered and more portable than a desktop computer.
- **Tablet Computer:** a handheld computer that is more portable than a laptop. Instead of a keyboard and mouse, tablets use a touch-sensitive screen for typing and navigation.
- **Smartphone:** a more powerful version of a traditional cell phone with a touch-sensitive screen. Smartphones can connect to the Internet and you can use them for browsing the Web, receiving and sending emails, or playing games.



Desktop



Laptop



Tablet



Smartphone

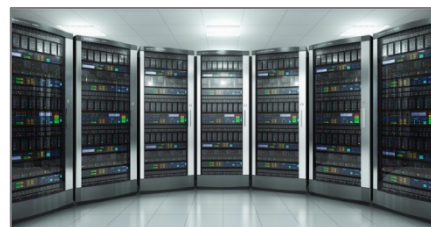
1.2.2 Server

A server is a software or hardware device that accepts and responds to requests made over a network. The device that makes the request, and receives a response from the server, is called *a client*. On the Internet, the term "Server" commonly refers to the computer system that receives requests for web files and sends those files to the client.



1.2.3 Mainframe Computer

A very large and expensive computer capable of supporting hundreds, or even thousands, of users simultaneously. In some ways, mainframes are more powerful than supercomputers because they support more simultaneous programs. However, supercomputers can execute a single program faster than a mainframe.



1.2.4 Supercomputer

The fastest and most powerful type of computer. Supercomputers are very expensive and are employed for specialized applications that require immense amounts of mathematical calculations, such as weather forecasting, nuclear energy research, and petroleum exploration.

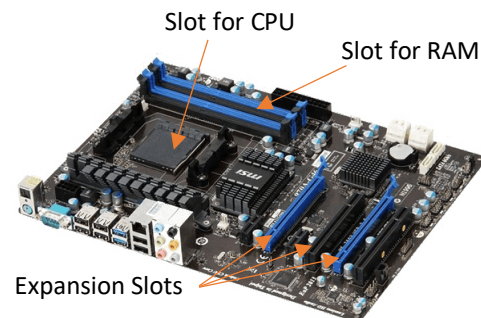


1.3 HARDWARE

The term hardware refers to the physical component of the computer. Computer hardware includes input devices, output devices, central processing unit (CPU), motherboard, power supply, random access memory (RAM), and other components.

1.3.1 Motherboard

The motherboard is the main circuit board of your computer. It is where most of the parts and peripherals are connected. It holds the central processing unit CPU, random access memory RAM, read only memory ROM, and other components.



1.3.2 The Central Processing Unit (CPU)

Central processing unit (CPU) is the central component of the Computer System. Sometimes it is called as microprocessor or processor. It is the brain of the computer. All functions and processes that are done on a computer are performed directly or indirectly by the processor.



The CPU consists of transistors that receives data and instructions, process (calculate) the data according to the instructions, and produce the results of the calculations. The most common types of computer processors are Intel and AMD.

❖ Processor Speed

The Speed of the CPU is a primary factor that determines the computer's performance. CPU speed is the number of cycles per second at which the processor operates and can process information. The more cycles that a processor unit can complete per second, the faster data can be processed. Processor speed is measured in *megahertz*.

A hertz equals one cycle per second. For Example, a processor with 3.8 GHz runs 3.8 billion cycle per second. To help CPUs run faster, multi-core processors were developed. Multi-core processor means having two or more processors on one integrated circuit, such as: dual-core and quad-core processors.

❖ Parts of the processor:

1. **Arithmetic Logic Unit (ALU):** It is the part of the computer processor (CPU) used to perform arithmetic operations (such as: addition, subtraction, multiplication, and division) and logic operations such as comparison.
2. **Control Unit (CU):** It directs operations within a computer's processor. It receives instructions from a program, then passes them to the arithmetic logic unit (ALU), and sends these results of processing to the corresponding program as output.
3. **Register:** It is a temporary storage area of the computer processor. It holds data that is being worked on by the processor. The size of the register is measured in *bits*. The available sizes in the market are 32-bit or 64-bit.

❖ How a processor works?

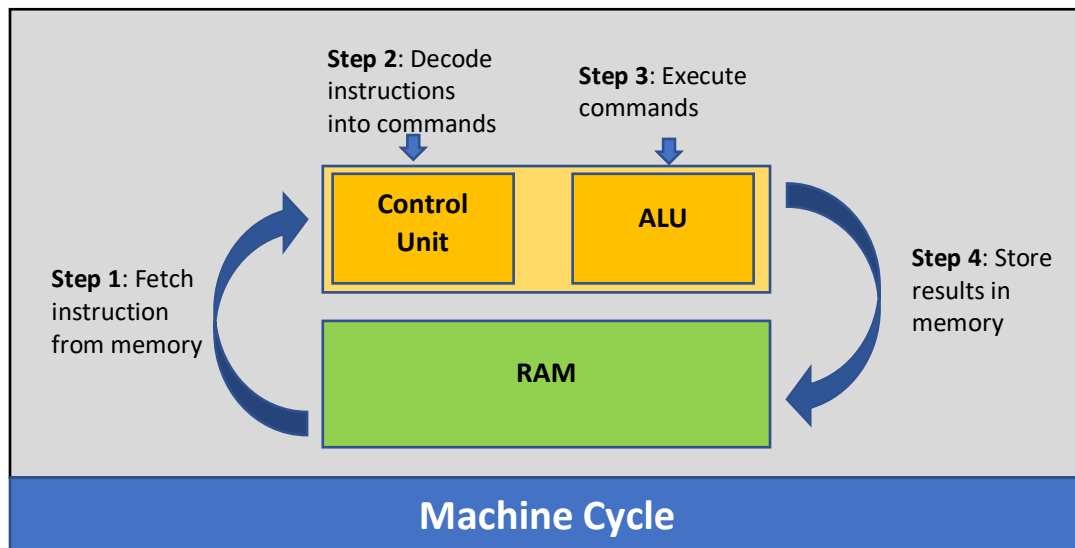
For every single instruction to be execute, the CPU repeats the machine cycle that consists of four operations: *fetching, decoding, exulting, and storing*.

Fetching: is the process of getting the instruction or data from the memory.

Decoding: is the process of translating the instruction into signals that computer can execute.

Executing: is the process of carrying out the command.

Storing: means writing the result to the memory.



Lec 2: COMPUTER FUNDAMENTALS- Part1

2.1 Memory

It is also called the primary memory. It consists of electronic chips that holds: a) the operating system and other system software, b) the application programs, and c) the data being processed and resulting information.

2.1.1 RAM (Random Access Memory)

The **RAM** is the main memory where the operating system is loaded and also where your applications are copied to, when you load an application. The capacity of RAM plays an important role in the performance of the computer, the more capacity the RAM has, the more programs you can open at the same time. If your system is running slowly due to a lack of RAM, it is better to add more RAM modules to increase its capacity.



The CPU directly accesses the content of the RAM. The storage of data (or information) and instructions on RAM are temporary. Therefore, we can say that RAM is a volatile memory. When the power is switched off the RAM becomes empty.

2.1.2 ROM (Read-Only Memory): is a special chip held on your computer's system (motherboard) that can be read only (not changed). It contains software that is required to make your computer work with your operating system. The content in the ROM is permanent, so ROM is a non-volatile memory.



2.2 Input Devices

Input devices: any peripheral (a piece of computer hardware equipment) used to provide data and send instructions to the computer.

1. **Keyboard:** a device to input text and characters by pressing buttons (referred to as keys), similar to a typewriter. The most common English-language key layout is the QWERTY layout. It has around 104 keys, including alphabetic characters, punctuation symbols, numbers, and a variety of function keys



2. Pointing Devices

- **Optical Mouse:** uses lasers, or more commonly LEDs, to track the surface under the mouse to determine the motion of the mouse, to be translated into mouse movements on the screen.
- **Trackball:** a pointing device consisting of an exposed extended ball housed in a socket that detects rotation about two axes.
- **Touchscreen:** It is a computer screen that can be used by touching it with a finger or a stylus pen, instead of using a mouse and keyboard. Touchscreens are used in Personal Digital Assistants (PDAs), tablet computers, smartphones and other devices



3. **Game controller** – an input device used with video games or entertainment systems to provide input to a video game. Game controllers could include gamepads, joysticks, steering wheels, paddle, and the keyboard and mouse.



Gamepad



Joystick



Steering wheel and paddle

4. **Image Scanner** - a device that provides input by analyzing images, printed text, handwriting, or an object. Scanners capture images from the paper document and convert them into a digital format.



5. **Webcam**: a low-resolution video camera used to provide visual input that can be easily transferred over the internet.



6. **Digital Camera**: It is used to capture pictures or video which is then stored into its memory card. Stored digital images or video can be transfer to a computer by connecting the camera or the memory card to it.



7. **Fingerprint scanner** - is a type of technology that identifies and authenticates the fingerprints of an individual in order to grant or deny access to a computer system or a physical facility.



8. **Microphone**: A sound sensor that provides input by converting sound into electrical signals.



9. **Bar code reader or Universal Product Code (UPC)** - a hardware input device capable of reading a barcode using a laser. An example of a barcode reader is a supermarket barcode scanner that reads and logs the price of a product.



10. **Magnetic Ink Character Recognition (MICR)** - a character-distinguishing technology that makes use of special magnetized ink. It is largely used in banks and other organizations where security is a high priority.



11. Optical Character Reader (OCR) - is used for taking an image of letters or typed text and converting it into data the computer understands.



12. QR (Quick Response) Code - is a two-dimensional barcode with a larger storage capacity than the standard UPC (i.e. barcode). A QR code reader app can be installed on a smartphone, allowing the user to scan a QR code and view the data stored in it. QR codes often contain information about a product or a direct forward to a website.



2.3 Output Devices

Output Device: is any piece of computer hardware used to communicate the results of data processing to the user.

1. **Monitor or Visual Display Unit (VDU):** it is used for outputting information in an understandable format for humans. Older computer monitors made use of Cathode Ray Tubes (**CRT**), which made them large, heavy and inefficient. Nowadays, flat-screen **LCD** (Liquid Crystal Display) or **LED** (Light Emitting Diode) monitors are lighter and more energy efficient.



2. **Printers:** an output device that are used to print information on paper (referred as hard-copy) There are many different types of printers such as Dot-Matrix printer, Ink-Jet printer and Laser printer. In large organizations, laser printers are most commonly used because they can print very fast and give a very high quality output.



Dot-Matrix Printer



Ink-Jet Printer



Laser Printer

3. **Plotters:** it is an output device similar to a printer, but normally allows you to print larger images.



4. **3D Printer:** is a device that creates a physical object from a digital model by layering materials. 3D printers use materials, such as metal alloys, polymers, plastics, or even food ingredients.

5. **Speakers:** an output hardware device that connects to a computer to generate sound.



6. **A projector or image projector:** It is an optical device, which projects an image (or videos) onto a surface, commonly a projection screen. The newest types of projectors are handheld projectors that use lasers or LEDs to project images.

